Mini Report: Unsupervised Refinement on Qwen3 Audio Embeddings

NXSCough Experiments

March 19, 2026

Pipeline Summary

1. Extract embeddings with `Qwen3_Extractor` (mean+std pooled representation).
2. Baseline supervised evaluation across 3 classifiers.
3. Harmonize embedding space with empirical Bayes ComBat.
4. Detect/remove outliers (Isolation Forest + per-database centroid distance ensemble, 3%).
5. Estimate acoustic TB likelihood (outlier-factor score + Bayesian GMM posterior ensemble).
6. Relabel TB samples into:
   - **Acoustic-TB** (high score),
   - **Label-only-TB** (low score), then filter label-only TB.

Qwen3 Extractor Architecture

- Backbone: audio tower of `Qwen3-Omni-30B-A3B-Thinking`, used as a **frozen feature extractor**.
- Audio pipeline:

  wav $\rightarrow$ Mel frontend $\rightarrow$ Qwen3 audio encoder ($3\times$ CNN stride) $\rightarrow$ frame embeddings

- The encoder outputs frame-level hidden states with base dimension **2048**.
- Temporal pooling: compute mean μ and standard deviation σ over the time axis.
- Final utterance embedding:

$$z = [\mu\,;\,\sigma] \in \mathbb{R}^{4096}$$

  Concatenating mean and std doubles the dimension to **4096**, which is used for all downstream experiments.

Goal

Extract a fixed-length, TB-relevant audio representation from raw waveforms without any fine-tuning.

Classifier Evaluation Protocol

- Three classifiers are compared on the same embedding space:
  - Logistic Regression, MLP, SVM (RBF).
- Each classifier is wrapped in a pipeline:

$$\text{StandardScaler} \rightarrow \text{PCA}(50) \rightarrow \text{Classifier}$$

- Cross-validation uses `StratifiedGroupKFold` with participant ID as group key, ensuring no patient appears in both train and validation folds simultaneously.
- Training is done on CODA (DB0); external evaluation is done on CIRDZ (DB1) and **TBScreen** (DB2).
- Final reported score: mean AUROC across folds, then averaged across all classifiers.

Goal

Evaluate TB detection quality at each pipeline stage using a consistent, leakage-free protocol across multiple classifiers.

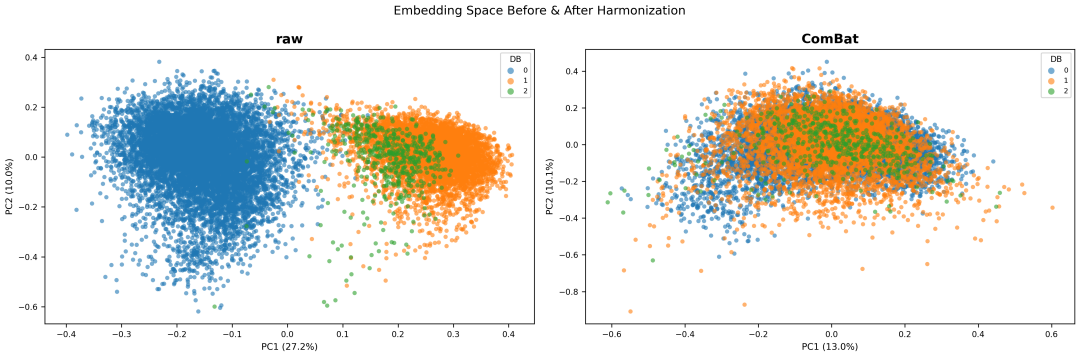
ComBat Harmonization

- ComBat is an empirical Bayes batch-effect correction method originally developed for genomics.
- Each database is treated as one batch. It estimates and removes:
  - additive batch shift $\hat{\gamma}$ (location),
  - multiplicative batch scale $\hat{\delta}$ (variance).
- Harmonization quality is evaluated with Ridge-based predictability of database labels in PCA space:
  - Raw: $R^2_{\text{db}} = 0.0198$, $R^2_{\text{dis}} = 0.0023$
  - ComBat: $R^2_{\text{db}} \approx 2.16 \times 10^{-8}$, $R^2_{\text{dis}} = 0.0017$
- After ComBat, database identity (CODA vs. CIRDZ) is no longer predictable from the embedding space, while disease information is partially preserved.

Purpose

Reduce dataset-specific (batch) variation in embeddings so that downstream classifiers focus on TB-related acoustic structure rather than database identity.

ComBat Result



Caption: PCA projections of embeddings before (left) and after (right) ComBat harmonization. Database-related variation is reduced after correction. External AUROC improves on CIRDZ from 0.50 ± 0.05 to 0.55 ± 0.02 and on TBScreen from 0.47 ± 0.04 to 0.48 ± 0.02.

Outlier Detection Strategy

- Assumed contamination rate: **3%**.
- Two complementary scores are combined:
  - **Isolation Forest**: detects globally anomalous points in PCA-reduced (50-dim) embedding space.
  - **Per-database centroid distance**: flags samples whose distance from the database centroid exceeds $\sigma_{\text{threshold}} = 2.5$ standard deviations, normalized by Median Absolute Deviation.
- Weighted ensemble outlier score:

$$s = 0.50\, s_{\text{IF}} + 0.50\, s_{\text{centroid}}$$

- Samples in the top 3% of s are removed before the next stage.

Purpose

Remove samples whose embeddings are unlikely to represent clean cough signals, such as noisy recordings or non-cough acoustic content.

Outlier Removal Result

Caption: Left: detected outliers ($\times$) overlaid on PCA projection. Right: retained inlier cloud after cleaning. Removed the top **3%** by ensemble outlier score. Mean AUROC across classifiers on CODA remains stable (0.66 ± 0.03) while external AUROC improves on TBScreen to 0.53 ± 0.02.

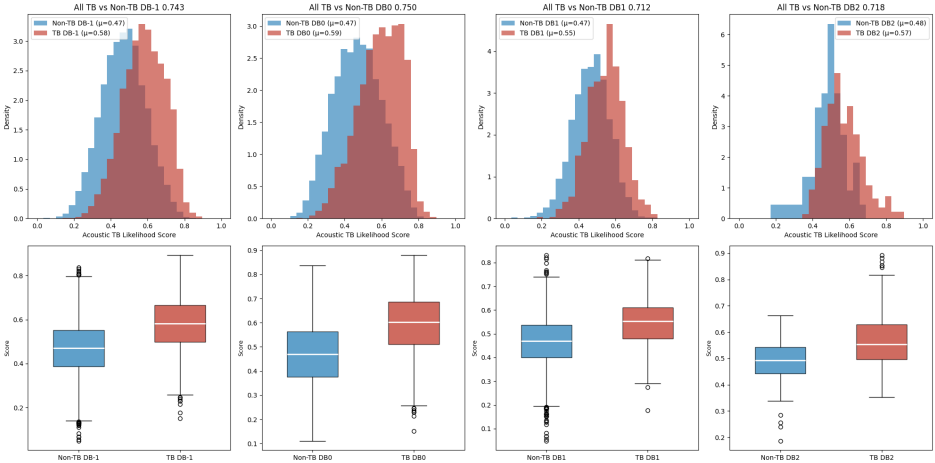
Acoustic TB Likelihood Modeling

- **Outlier-factor margin score (LOF)** — a density-based "TB-likeness" score:
  - Fit Local Outlier Factor models on TB and Non-TB subsets in PCA space.
  - Compute a posterior-like score from the relative class densities (high score $\Rightarrow$ more TB-like).
- **Bayesian GMM posterior (BGMM)** — class-conditional density models with a Dirichlet-process prior:
  - Fit separate Bayesian Gaussian mixture models on TB and Non-TB embeddings.
  - Score $= P(\mathrm{TB} \mid x)$ via a numerically-stable posterior computed from log-likelihoods.
- Final acoustic TB likelihood: simple average of LOF-margin and BGMM scores.

Goal

Assign each sample an unsupervised score reflecting how acoustically TB-like it is in the embedding space, to enable label refinement without additional annotations.

Acoustic TB Likelihood Result



Caption: Top: score distributions for TB and Non-TB. Bottom: box plots with Mann–Whitney U significance test. Scores are produced by the LOF+BGMM ensemble; TB scores are shifted higher than Non-TB scores, indicating separable acoustic structure.

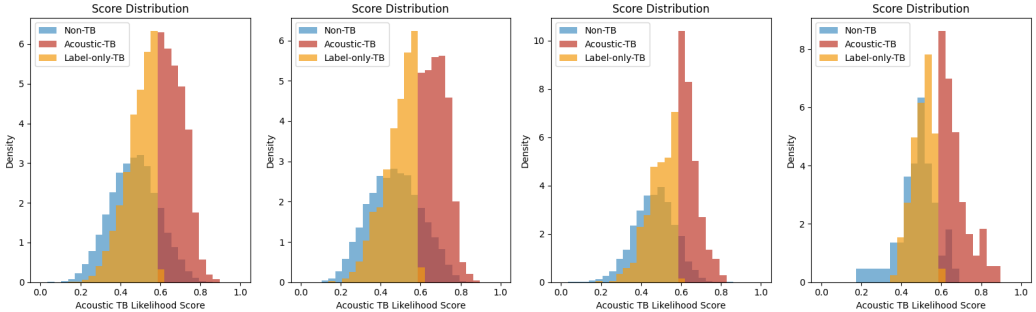
Relabeling Strategy

- Only original TB-labeled samples are relabeled; Non-TB samples are unchanged.
- Threshold τ is estimated from the *1D acoustic-score distribution* using a Gaussian mixture model:
  - Fit a GMM to all acoustic scores and select the best model by BIC.
  - Choose the highest-mean component as the **Acoustic-TB** mode.
  - Set τ at the posterior crossover point where $P(\text{Acoustic-TB} \mid s) = 0.5$.
  - Fallback: if the score distribution is unimodal, use the 95th percentile of Non-TB scores.
- Interpretation by position relative to τ:
  - score $\geq \tau \Rightarrow$ **Acoustic-TB**: embedding strongly matches the TB acoustic profile.
  - score $< \tau \Rightarrow$ **Label-only-TB**: annotated as TB but acoustically ambiguous; excluded from training.
- Optional refinement: participant-level majority voting to reduce segment-level noise.
- This is unsupervised label refinement, not class balancing.

Purpose

Refine noisy TB annotations by retaining only samples that are both annotated as TB *and* acoustically consistent with the TB embedding cluster.

Relabeling Result



Caption: Score distribution after relabeling. Left of the dashed threshold τ: Label-only-TB (acoustically ambiguous, excluded). Right of τ: Acoustic-TB (kept). The split is summarized per database (CODA/CIRDZ/TBScreen) to verify that the refinement is not dominated by a single source.

Classification Performance (AUROC)

Stage	CODA (CV)	CIRDZ (Ext.)	TBScreen (Ext.)
Baseline	0.66 ± 0.03	0.50 ± 0.05	0.47 ± 0.04
After ComBat	0.67 ± 0.03	0.55 ± 0.02	0.48 ± 0.02
After Outlier Removal	0.66 ± 0.03	0.55 ± 0.02	0.53 ± 0.02
After Acoustic-TB Filtering	0.87 ± 0.01	0.71 ± 0.01	0.88 ± 0.02

Summary: ComBat and outlier removal provide incremental gains.
Acoustic-TB filtering is the dominant step, improving external generalization on both CIRDZ (DB1) and TB-Screen (DB2).